

LINUX TEST

Operating Systems / Linux Fundamentals

Total Marks	100	Duration	3 Hours
Name	Tavakalmastan	Domain / Date	PD/16-06-2026

General Instructions

1. This paper has three sections: A (MCQs), B (Short Answer) and C (Practical/Lab Tasks).
2. All questions are compulsory.
3. Section A carries 1 mark per question; Section B carries 2 marks per question.
4. Section C (Practical) is evaluated out of 47 marks based on the rubric for each task — execute the required commands and paste screenshots showing both commands and outputs.
5. Use of mobile phones, books, or notes is strictly prohibited.

Test Submission Instructions

1. The question sheet will be provided in **Microsoft Word (.docx) format**.
2. **For Multiple Choice Questions (MCQs):**
 - Highlight the correct option using a suitable color (e.g., yellow).
 - Ensure the selected answer is clearly visible.
3. **For Short Answer Questions:**
 - Write your answer directly below the respective question.
 - Keep answers clear, concise, and relevant.
4. **For Scenario-Based Questions:**
 - Execute the solution in the terminal.
 - Take a screenshot showing both the command(s) and the output.
 - Insert the screenshot below the corresponding question.
5. **Final Submission:**
 - After completing all questions, save the document as a **PDF**.
 - Rename the PDF file using your full name in the format: YourName.pdf
6. **Submit the renamed PDF file before the specified deadline, Link will be provided to submit the test.**

Note: Ensure all screenshots are clear and readable. Incomplete or improperly named submissions may not be considered for evaluation.

Section A — Multiple Choice Questions ($1 \times 31 = 31$ Marks)

1. What does the command **chmod 754 file.txt** do?

A. **rwxr-xr--** B. **rwxr-xr-x** C. **rwxr-xr—x** D. **rwxr--r--**

2. Which command displays the inode number of files?

A. **ls -l** B. **ls -n** C. **stat -f** D. **find -inode**

3. What is the purpose of the command?

```
grep -r "error" /var/log
```

- A. Search for "error" only in files directly under /var/log
- B. Search recursively for "error" in all files under /var/log**
- C. Remove lines containing "error"
- D. Count occurrences of "error"

4. What will this command do?

```
find . -type f -mtime -7
```

- A. Find files modified exactly 7 days ago
- B. Find files modified within the last 7 days**
- C. Find files older than 7 days
- D. Find files larger than 7 MB

5. Which command can be used to display the last 50 lines of a log file?

- A. head -50 logfile
- B. tail -50 logfile**
- C. cat -50 logfile
- D. less +50 logfile

6. What is the output of the following command?

```
echo $?
```

- A. Current PID
- B. Previous command's exit status**
- C. User ID
- D. Shell version

7. Which command shows disk usage of directories in human-readable format?

- A. df -h
- B. du -sh ***
- C. ls -lh
- D. free -h

8. Which command finds all files owned by user "john"?

- A. find / -owner john
- B. find / -user john**
- C. ls -user john
- D. grep john /etc/passwd

9. Which command can find files larger than 100 MB?

```
find / -type f ____
```

- A. -size +100M**
- B. -size 100MB
- C. -larger 100M
- D. -gt 100M

10. Which command displays the command history with line numbers?

- A. history
- B. history -n
- C. history | nl
- D. Both A and C**

11. What does the command cp file1 file2 do?

- A. Deletes file1
- B. Copies file1 to file2**
- C. Renames file1
- D. Moves file1

12. Which command is used to move or rename a file?

- A. mv**
- B. cp
- C. rm
- D. cat

13. Which command removes a file?

- A. del B. erase **C. rm** D. remove

14. Which command shows the current logged-in user?

- A. user B. id **C. whoami** D. login

15. Which command displays the contents of a directory in long format?

- A. ls -a **B. ls -l** C. ls -h D. ls -r

16. Which command is used to search for text inside a file?

- A. grep** B. find C. locate D. search

17. Which command creates an empty file named notes.txt?

- A. mkdir notes.txt **B. touch notes.txt** C. cat notes.txt D. nano notes.txt

18. What will cp file1 file2 do if file2 already exists?

- A. Show an error B. Append data to file2
C. Overwrite file2 with file1's contents D. Delete file1

19. What does mv file1 file2 do when file2 does not exist?

- A. Copies file1 to file2 **B. Renames file1 to file2**
C. Deletes file1 D. Creates a directory

20. Which command shows the first 10 lines of a file by default?

- A. top B. tail **C. head** D. cat

21. Which command is used to find files by name?

- A. grep **B. find** C. search D. locateuser

22. Which command creates a directory named project?

- A. touch project **B. mkdir project**
C. mk project D. new project

23. What is the purpose of grep "error" log.txt?

- A. Delete lines containing "error" **B. Search for lines containing "error"**
C. Rename log.txt D. Count words

24. What does clear do?

- A. Deletes files from current directory **B. Clears terminal screen only**
C. Clears command history D. Restarts terminal

25. What does `cd` do when used without any arguments?

- A. Goes to the root directory (`/`) B. Stays in the current directory
C. Goes to the home directory D. Opens a new terminal

26. Which command is most suitable for viewing a very large text file page by page?

- A. `cat` B. `less` C. `rm` D. `pwd`

27. What does `rm -f file.txt` do?

- A. Forces deletion without confirmation messages B. Moves the file to Trash
C. Compresses the file D. Renames the file

28. Which command displays the amount of free and used memory?

- A. `du` B. `df` C. `free` D. `topdir`

29. Which command is used to change a user's password?

- A. `password` B. `passwd` C. `pass` D. `pwd`

30. What does `wc -l file.txt` count?

- A. Characters B. Words C. Lines D. Files

31. Suppose the current directory contains:

file1.txt
file2.txt

What will this command do?

`rm *.txt`

- A. Remove only file1.txt B. Remove only file2.txt
C. Remove all .txt files in the current directory D. Remove the directory itself

Section B — Short Answer Questions (2 × 11 = 22 Marks)

1. Explain the purpose of the inode in Linux.

ANS: An inode is a data structure that stores metadata about a file, such as ownership, permissions, size, timestamps and disk location

2. How can you find the top 10 largest files in a filesystem?

ANS: `find / -type f -exec du -h {} + 2>/dev/null | sort -rh | head -10`

3. How do you change ownership of a file or directory?

ANS: `sudo chown username filename`

4. Write a command to find all .log files modified in the last 7 days.

ANS: `find / -type f -name "*.log" -mtime -7`

5. What is the root user?

ANS: The root user is the user in Linux with unrestricted access to all files, commands and system settings

6. What is a shell?

ANS: A shell is a command line interpreter that allows users to interact with the operating system by executing commands

7. How do you move to the parent directory?

ANS: cd../

8. How do you move directly to your home directory?

ANS: cd

9. How do you remove an empty directory?

ANS: rmdir directory_name

10. Why are file permissions important?

ANS: File permissions control who can read, write or execute files, helping protect data and maintain system security

11. What command shows the current date and time?

ANS: date

Section C — Practical / Lab-Based Tasks *(Total 47 Marks)*

Instructions:

Execute the required Linux commands in the terminal and paste screenshots showing both and outputs.

Question 1: Employee Records Management

Perform the following tasks commands related to employee records management:

1. Create a directory named employees.
2. Create files emp1.txt, emp2.txt and emp3.txt.
3. List all files including hidden files.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux
$ mkdir employees

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux
$ cd employees/

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ touch emp1.txt emp2.txt emp3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ ls -l
total 0
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp1.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp2.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ ls -la
total 8
drwxrwx----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 .
drwxrwx----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:31 ..
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp1.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp2.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$
```

Question 2: Hidden Files Investigation

Perform the following tasks related to hidden files investigation:

1. Create a hidden file named .secret.
2. Display all files including hidden files.
3. Verify that the hidden file exists.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ touch .secret

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ ls -la
total 8
drwxrwx---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:36 .
drwxrwx---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:31 ..
-rwxrw---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:36 .secret
-rwxrw---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp1.txt
-rwxrw---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp2.txt
-rwxrw---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:32 emp3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ ls -l .secret
-rwxrw---+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 10:36 .secret

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$
```

Empowering Silicon Minds

Question 3: Archive Creation

Perform the following tasks related to archive creation:

1. Create a directory named project.
2. Create three files inside it.
3. Create a tar archive named project.tar.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ mkdir project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ cd project/

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ touch file1.txt file2.txt file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls
file1.txt  file2.txt  file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cd ..

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$ tar -czvf project.tar project
project/
project/file1.txt
project/file2.txt
project/file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees
$
```

Question 4: Compressed Backup

Perform the following tasks related to compressed backup:

1. Create a directory named backup_data.
2. Add files inside it.
3. Create a compressed archive backup.tar.gz.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ mkdir backup_data

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cd backup_data/

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/backup_data
$ touch file1.txt file2.txt file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/backup_data
$ ls
file1.txt  file2.txt  file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/backup_data
$ cd ..

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ tar -czvf backup.tar.gz backup_data/
backup_data/
backup_data/file1.txt
backup_data/file2.txt
backup_data/file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 5: Archive Extraction

Perform the following tasks related to archive extraction:

1. Extract backup.tar.gz.
2. Verify the extracted files and directories.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ rm -r backup_data/

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls
backup.tar.gz  file1.txt  file2.txt  file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ tar -xzvf backup.tar.gz
backup_data/
backup_data/file1.txt
backup_data/file2.txt
backup_data/file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls -R backup_data/
backup_data/:
file1.txt  file2.txt  file3.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$
```

Question 6: Process Monitoring

Perform the following tasks related to process monitoring:

1. Display currently running processes.
2. Display top resource-consuming processes.

Submit: Screenshot showing commands executed and corresponding outputs.

 /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ps -ef
  UID      PID  PPID  TTY      STIME  COMMAND
Tavakalm  1234    1233  pts/0    22:19:57 -bash
Tavakalm  1233        1  ?        22:19:57 /usr/bin/mintty -i /cy
Tavakalm  1708    1234  pts/0    11:09:36 ps -ef

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ps -w
  PID  PPID  PGID  WINPID  TTY      UID    STIME  C
4194308      0      0        4  ?        0      Jun 15 S
4194540      0      0       236  ?        0      Jun 15 *
4194584      0      0       280  ?        0      Jun 15 R
4195100      0      0       796  ?        0      Jun 15 C
4195556      0      0      1252  ?        0      Jun 15 C
4195668      0      0      1364  ?        0      Jun 15 C
4195676      0      0      1372  ?        0      Jun 15 C
4195744      0      0      1440  ?        0      Jun 15 C
4195756      0      0      1452  ?        0      Jun 15 C
4195772      0      0      1468  ?        0      Jun 15 C
4195836      0      0      1532  ?        0      Jun 15 C
4195964      0      0      1660  ?        0      Jun 15 C
4196004      0      0      1700  ?        0      Jun 15 C
4196012      0      0      1708  ?        0      Jun 15 C
4196088      0      0      1784  ?        0      Jun 15 C
4196140      0      0      1836  ?        0      Jun 15 C
4196192      0      0      1888  ?        0      Jun 15 C
4196256      0      0      1952  ?        0      Jun 15 C
4196336      0      0      2032  ?        0      Jun 15 C
4195740      0      0      1436  ?        0      Jun 15 C
4196188      0      0      1884  ?        0      Jun 15 C
4196376      0      0      2072  ?        0      Jun 15 C
4196388      0      0      2084  ?        0      Jun 15 C
4196424      0      0      2120  ?        0      Jun 15 C
4196644      0      0      2340  ?        0      Jun 15 C
4196660      0      0      2356  ?        0      Jun 15 C
4196684      0      0      2380  ?        0      Jun 15 C
4196736      0      0      2432  ?        0      Jun 15 C
4196748      0      0      2444  ?        0      Jun 15 C
4196760      0      0      2456  ?        0      Jun 15 C
4196832      0      0      2528  ?        0      Jun 15 C
4196916      0      0      2612  ?        0      Jun 15 C
4196924      0      0      2620  ?        0      Jun 15 C
4196980      0      0      2676  ?        0      Jun 15 C
4197048      0      0      2744  ?        0      Jun 15 C
4197116      0      0      2812  ?        0      Jun 15 C
```

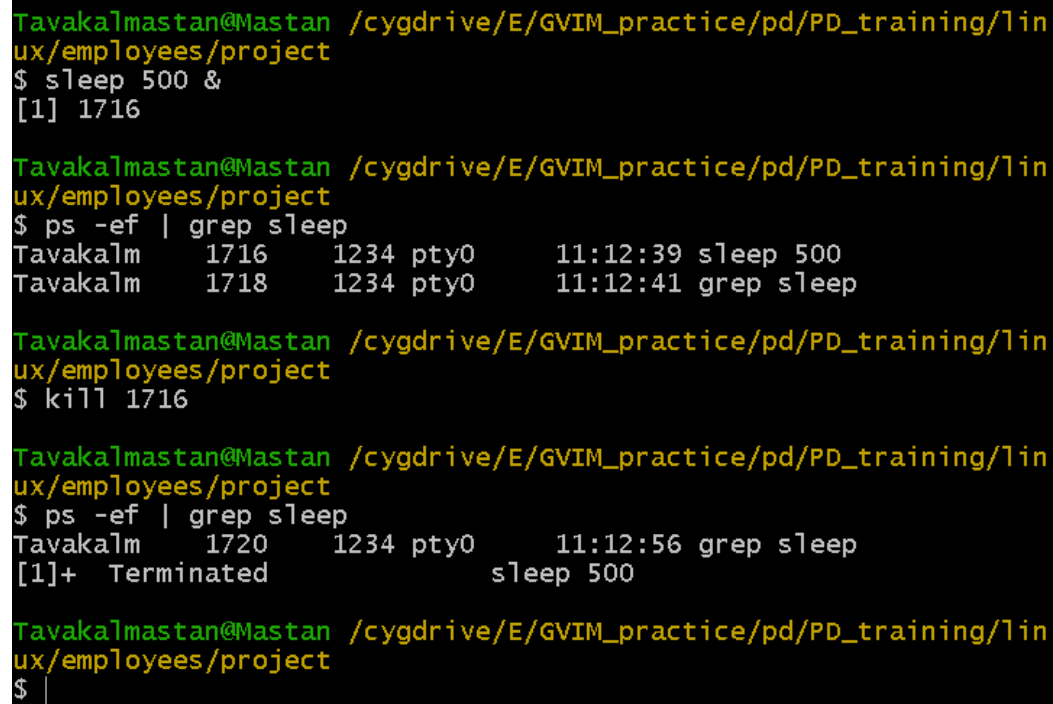
Question 7: Process Termination

Perform the following tasks related to process termination:

1. Start a background process using sleep 500.
2. Find its PID.
3. Kill the process.

4. Verify termination.

Submit: Screenshot showing commands executed and corresponding outputs.



```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ sleep 500 &
[1] 1716

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ps -ef | grep sleep
Tavakalm      1716      1234 pty0      11:12:39 sleep 500
Tavakalm      1718      1234 pty0      11:12:41 grep sleep

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ kill 1716

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ps -ef | grep sleep
Tavakalm      1720      1234 pty0      11:12:56 grep sleep
[1]+  Terminated                  sleep 500

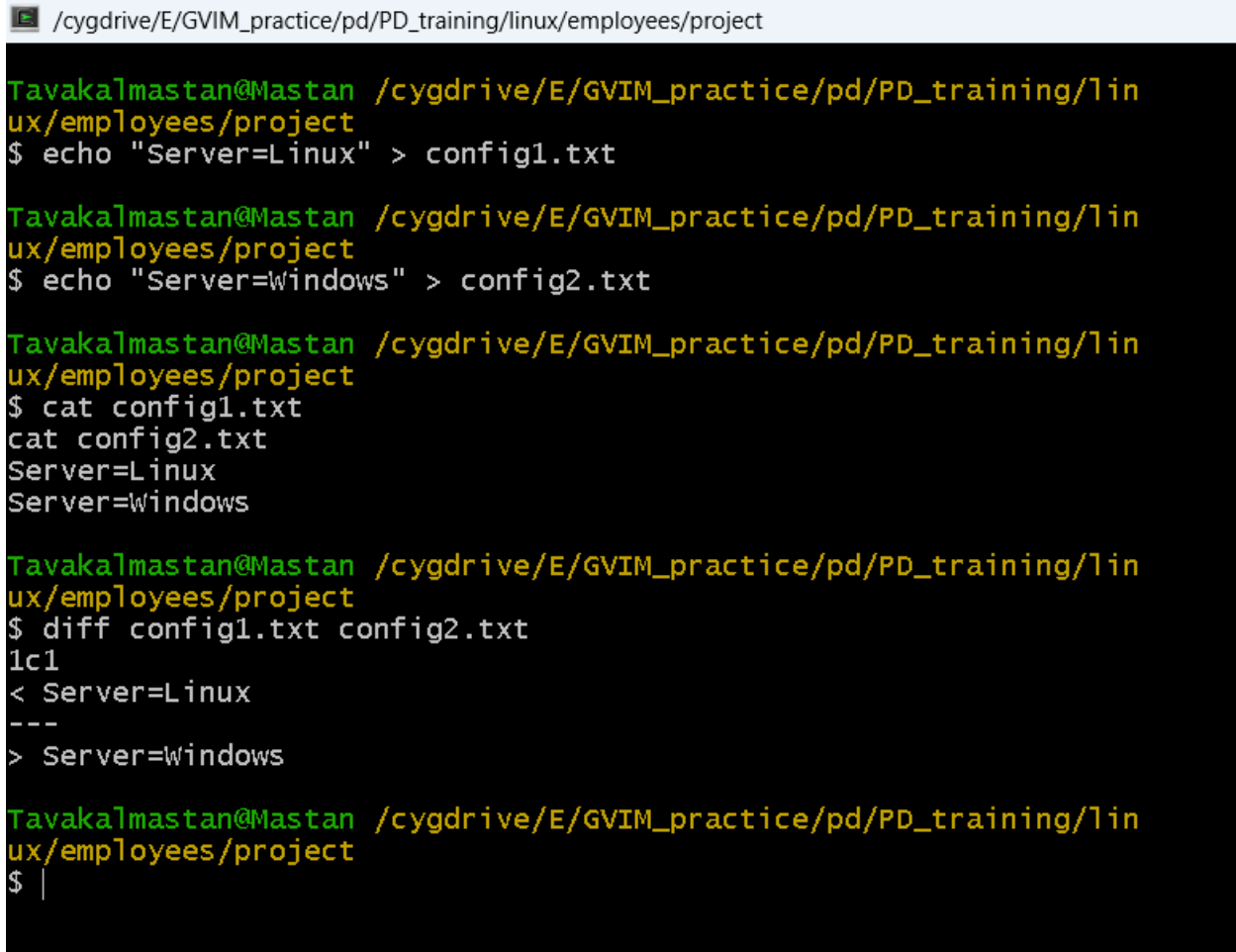
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 8: File Comparison

Perform the following tasks related to file comparison:

1. Create config1.txt and config2.txt.
2. Add different contents.
3. Compare both files.

Submit: Screenshot showing commands executed and corresponding outputs.



```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ echo "Server=Linux" > config1.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ echo "Server=Windows" > config2.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat config1.txt
cat config2.txt
Server=Linux
Server=Windows

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ diff config1.txt config2.txt
1c1
< Server=Linux
---
> Server=Windows

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 9: Sorting Data

Perform the following tasks related to sorting data:

1. Create employees.txt.
2. Add at least 10 employee names.
3. Sort the file alphabetically.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

ravakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat > employees.txt
sudeep
samarth
rishikesh
shridhar
vinod
srikar
limbraj
suhas
prashant
varun
vishal
anand

ravakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat employees.txt
sudeep
samarth
rishikesh
shridhar
vinod
srikar
limbraj
suhas
prashant
varun
vishal
anand

ravakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ sort employees.txt
anand
limbraj
prashant
rishikesh
samarth
shridhar
srikar
sudeep
suhas
varun
vinod
vishal
```

Question 10: Duplicate Removal

Perform the following tasks related to duplicate removal:

1. Create students.txt.
2. Add duplicate entries.
3. Display only unique entries.

Submit: Screenshot showing commands executed and corresponding outputs.

 /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat > student.txt
sagar
shivaraj
prajwal
dhanraj
suhas
shivaraj
sagar
shridhar

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ sort student.txt | uniq

dhanraj
prajwal
sagar
shivaraj
shridhar
suhas

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 11: Network Information

Perform the following tasks related to network information:

1. Display IP address information.
2. Display hostname.
3. Display routing table.

Submit: Screenshot showing commands executed and corresponding outputs.

 /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
```

```
$ ip addr
```

```
-bash: ip: command not found
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
```

```
$ ipconfig
```

```
Windows IP Configuration
```

```
Ethernet adapter Ethernet 4:
```

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

```
Ethernet adapter Ethernet 2:
```

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::66d2:7d85:acf1:62b0%23
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

```
Wireless LAN adapter Local Area Connection* 1:
```

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

```
Wireless LAN adapter Local Area Connection* 2:
```

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
```

```
Ethernet adapter VMware Network Adapter VMnet1:
```

```
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::f786:401f:4961:6582%7
IPv4 Address. . . . . : 192.168.157.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
```

```
Ethernet adapter VMware Network Adapter VMnet8:
```

```
Connection-specific DNS Suffix . :
```



```

/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
Connection-specific DNS Suffix  . :
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ hostname
Mastan

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ route print
=====
Interface List
 12...00 ff 9a 27 2e 8d .....Sophos TAP Adapter
 23...0a 00 27 00 00 17 .....VirtualBox Host-Only Ethernet Adapter
 20...74 13 ea ed e6 4a .....Microsoft Wi-Fi Direct Virtual Adapter
 4...76 13 ea ed e6 49 .....Microsoft Wi-Fi Direct Virtual Adapter #2
 7...00 50 56 c0 00 01 .....VMware Virtual Ethernet Adapter for VMnet1
 17...00 50 56 c0 00 08 .....VMware Virtual Ethernet Adapter for VMnet8
 14...74 13 ea ed e6 49 .....Intel(R) Wi-Fi 6 AX201 160MHz
 19...04 bf 1b 87 07 c6 .....Realtek PCIe GbE Family Controller
 1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway          Interface
Metric
 0.0.0.0                    0.0.0.0          10.118.249.252   10.118.249.
141 50
 10.118.249.0              255.255.255.0    On-link         10.118.249.
141 306
 10.118.249.141            255.255.255.255  On-link         10.118.249.
141 306
 10.118.249.255            255.255.255.255  On-link         10.118.249.
141 306
 127.0.0.0                  255.0.0.0        On-link         127.0.
0.1 331
 127.0.0.1                  255.255.255.255  On-link         127.0.
0.1 331
 127.255.255.255           255.255.255.255  On-link         127.0.

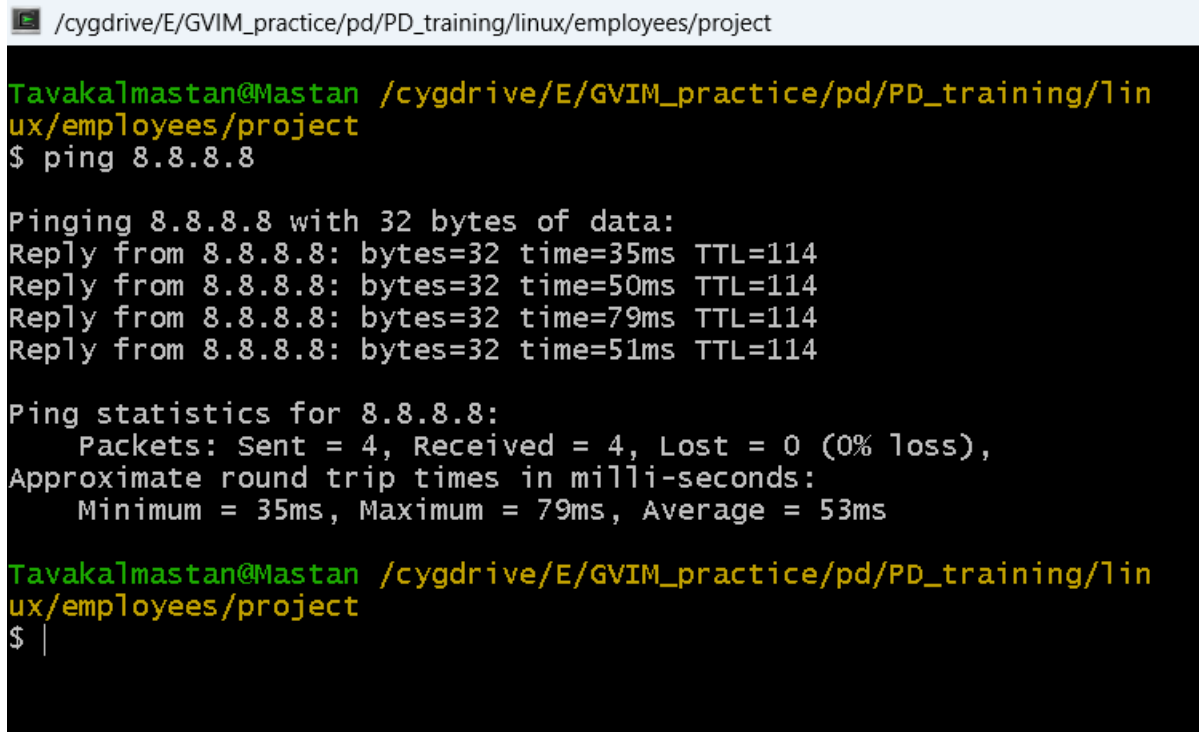
```

Question 12: Connectivity Test

Perform the following tasks related to connectivity test:

1. Verify connectivity to Google DNS.
2. Display the result of the connectivity test.

Submit: Screenshot showing commands executed and corresponding outputs.



```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project  
  
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project  
$ ping 8.8.8.8  
  
Pinging 8.8.8.8 with 32 bytes of data:  
Reply from 8.8.8.8: bytes=32 time=35ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=50ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=79ms TTL=114  
Reply from 8.8.8.8: bytes=32 time=51ms TTL=114  
  
Ping statistics for 8.8.8.8:  
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
    Approximate round trip times in milli-seconds:  
        Minimum = 35ms, Maximum = 79ms, Average = 53ms  
  
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project  
$ |
```

Question 13: User Account Information

Perform the following tasks related to user account information:

1. Display current username.
2. Display UID and GID information.
3. Display currently logged-in users.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ whoami
Tavakalmastan

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ id
uid=197609(Tavakalmastan) gid=197609(Tavakalmastan) groups=197609(Tavakalmastan),401408(Medium Mandatory Level),545(Users),4(INTERACTIVE),66049(CONSOLE LOGON),11(Authenticated Users),15(This Organization),68602(MicrosoftAccount+mastan1234t@gmail.com),113(Local account),4095(CurrentSession),66048(LOCAL),262180(Cloud Account Authentication)

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ who

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 14: Environment Variables

Perform the following tasks related to environment variables:

1. Display HOME directory path.
2. Display PATH variable.
3. Save both outputs to a file.

Submit: Screenshot showing commands executed and corresponding outputs.

 /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ echo $HOME
/home/Tavakalmastan

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ echo $PATH
/usr/local/bin:/usr/bin:/cygdrive/c/Program Files/OpenSSH:/cygdrive/c/Program Files (x86)/VMware/VMware Workstation/bin:/cygdrive/c/windows/system32:/cygdrive/c/windows:/cygdrive/c/windows/System32/wbem:/cygdrive/c/windows/System32/WindowsPowerShell/v1.0:/cygdrive/c/windows/System32/OpenSSH:/cygdrive/c/Program Files/dotnet:/cygdrive/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/cygdrive/c/Program Files/NVIDIA Corporation/NVIDIA NVDLISR:/cygdrive/c/MingW/bin:/cygdrive/c/Program Files (x86)/IVI Foundation/VISA/WinNT/Bin:/cygdrive/c/Program Files/IVI Foundation/VISA/Win64/Bin:/cygdrive/c/Program Files (x86)/IVI Foundation/VISA/WinNT/Bin:/cygdrive/e/Program Files/MATLAB/R2024a/bin:/cygdrive/c/Program Files/Polyspace/R2021a/runtime/win64:/cygdrive/c/Program Files/Polyspace/R2021a/bin:/cygdrive/c/WINDOWS/system32:/cygdrive/c/WINDOWS:/cygdrive/c/WINDOWS/System32/wbem:/cygdrive/c/WINDOWS/System32/WindowsPowerShell/v1.0:/cygdrive/c/WINDOWS/System32/OpenSSH:/cygdrive/c/Program Files/Git/cmd:/cygdrive/c/Program Files/nodejs:/cygdrive/c/Program Files/MySQL/MySQL Shell 8.0/bin:/cygdrive/e/Users/Tavakalmastan/AppData/Local/Programs/Python/Python313/Scripts:/cygdrive/e/Users/Tavakalmastan/AppData/Local/Programs/Python/Python313:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Programs/oh-my-posh/bin:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Programs/Python/Python313/Scripts:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Programs/Python/Python313:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Microsoft/windowsApps:/cygdrive/c/MingW/bin:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Programs/Microsoft VS Code/bin:/cygdrive/c/intelFPGA/20.1/modelsim_ase/win32aloem:/cygdrive/c/Users/Tavakalmastan/AppData/Local/GitHubDesktop/bin:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Pandoc:/cygdrive/c/Users/Tavakalmastan/AppData/Roaming/Amazon:/cygdrive/c/Users/Tavakalmastan/AppData/Roaming/npm

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ echo $HOME > env.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ echo $PATH >> env.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
```

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
akalmastan/AppData/Roaming/Amazon:/cygdrive/c/Users/Tavakalmastan
/AppData/Roaming/npm

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/lin
ux/employees/project
$ echo $HOME > env.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/lin
ux/employees/project
$ echo $PATH >> env.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/lin
ux/employees/project
$ cat env.txt
/home/Tavakalmastan
/usr/local/bin:/usr/bin:/cygdrive/c/Program Files/OpenSSH:/cygdr
ive/c/Program Files (x86)/VMware/VMware Workstation/bin:/cygdrive/
c/windows/system32:/cygdrive/c/windows:/cygdrive/c/windows/System
32/wbem:/cygdrive/c/windows/System32/windowsPowerShell/v1.0:/cygd
rive/c/windows/System32/OpenSSH:/cygdrive/c/Program Files/dotnet:/
cygdrive/c/Program Files (x86)/NVIDIA Corporation/PhysX/Common:/
cygdrive/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/cygdr
ive/c/MinGW/bin:/cygdrive/c/Program Files (x86)/IVI Foundation/VI
SA/winNT/Bin:/cygdrive/c/Program Files/IVI Foundation/VISA/win64/
Bin:/cygdrive/c/Program Files (x86)/IVI Foundation/VISA/winNT/Bin
:/cygdrive/e/Program Files/MATLAB/R2024a/bin:/cygdrive/c/Program
Files/Polyspace/R2021a/runtime/win64:/cygdrive/c/Program Files/Pol
yspace/R2021a/bin:/cygdrive/c/WINDOWS/system32:/cygdrive/c/WINDO
WS:/cygdrive/c/WINDOWS/System32/wbem:/cygdrive/c/WINDOWS/System32
/windowsPowerShell/v1.0:/cygdrive/c/WINDOWS/System32/OpenSSH:/cyg
drive/c/Program Files/Git/cmd:/cygdrive/c/Program Files/nodejs:/c
ygdrive/c/Program Files/MySQL/MySQL Shell 8.0/bin:/cygdrive/e/Use
rs/Tavakalmastan/AppData/Local/Programs/Python/Python313/Scripts:/
cygdrive/e/Users/Tavakalmastan/AppData/Local/Programs/Python/Pyt
hon313:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Programs/oh-
my-posh/bin:/cygdrive/c/Users/Tavakalmastan/AppData/Local/Program
s/Python/Python313/Scripts:/cygdrive/c/Users/Tavakalmastan/AppDat
a/Local/Programs/Python/Python313:/cygdrive/c/Users/Tavakalmastan
/AppData/Local/Microsoft/WindowsApps:/cygdrive/c/MinGW/bin:/cygdr
ive/c/Users/Tavakalmastan/AppData/Local/Programs/Microsoft VS Cod
e/bin:/cygdrive/c/intelFPGA/20.1/modelsim_ase/win32aloem:/cygdriv
e/c/Users/Tavakalmastan/AppData/Local/GitHubDesktop/bin:/cygdrive
/c/Users/Tavakalmastan/AppData/Local/Pandoc:/cygdrive/c/Users/Tav
akalmastan/AppData/Roaming/Amazon:/cygdrive/c/Users/Tavakalmastan
/AppData/Roaming/npm

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/lin
ux/employees/project
$
```

Question 15: Redirection and Pipes

Perform the following tasks related to redirection and pipes:

1. Display directory listing.
2. Save output to report.txt.
3. Count number of lines in the report using a pipe.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls -l > report.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat report.txt
total 9
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 177 Jun 16 11:03 backup.tar.gz
drwxrwx---+ 1 Tavakalmastan Tavakalmastan  0 Jun 16 11:03 backup_data
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 13 Jun 16 11:14 config1.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 15 Jun 16 11:14 config2.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 90 Jun 16 11:16 employees.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 1926 Jun 16 11:27 env.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan  0 Jun 16 11:01 file1.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan  0 Jun 16 11:01 file2.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan  0 Jun 16 11:01 file3.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan  0 Jun 16 11:29 report.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 62 Jun 16 11:20 student.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat report.txt | wc -l
12

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$
```

Question 16: Word Count Analysis

Perform the following tasks related to word count analysis:

1. Create a text file with at least 15 lines.
2. Display number of lines, words and characters.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ seq 1 15 > textfile.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cat textfile.txt
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ wc textfile.txt
15 15 36 textfile.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$
```

Question 17: Locate Command

Perform the following tasks related to locate command:

1. Update the locate database.
2. Search for files named passwd.
3. Display the results.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ which locate
/usr/bin/locate

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ which updatedb
/usr/bin/updatedb

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ updatedb
/usr/bin/find: '/cygdrive/c/$Recycle.Bin/S-1-5-18': Permission denied
/usr/bin/find: '/cygdrive/c/$Recycle.Bin/S-1-5-21-2852919885-2853516565-582874725-1002': Permission denied
/usr/bin/find: '/cygdrive/c/$Recycle.Bin/S-1-5-21-2852919885-2853516565-582874725-500': Permission denied
/usr/bin/find: '/cygdrive/c/Config.Msi': Permission denied
/usr/bin/find: '/cygdrive/c/PerfLogs': Permission denied
/usr/bin/find: '/cygdrive/c/Program Files/National Instruments/Shared/Web Server/certs/keys': Permission denied

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ locate passwd

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 18: Symbolic Link Creation

Perform the following tasks related to symbolic link creation:

1. Create a file named project.txt.
2. Create a symbolic link named project_link.
3. Verify the symbolic link.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ touch project.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls -l project.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 11:40 project.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ln -s project.txt project_link

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls -l project_link
lrwxrwxrwx 1 Tavakalmastan Tavakalmastan 11 Jun 16 11:40 project_link -> project.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ |
```

Question 19: Directory Tree Analysis

Perform the following tasks related to directory tree analysis:

1. Create the directory structure company/HR, company/Finance and company/IT.
2. Display the directory tree structure.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls
backup.tar.gz  employees.txt  file3.txt      student.txt
backup_data    env.txt        project.txt     textfile.txt
config1.txt    file1.txt      project_link
config2.txt    file2.txt      report.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ mkdir -p company/HR company/Finance company/IT

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls company/
Finance HR IT

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ ls
backup.tar.gz  config2.txt  file2.txt      report.txt
backup_data    employees.txt  file3.txt      student.txt
company        env.txt       project.txt     textfile.txt
config1.txt    file1.txt     project_link

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project
$ cd company/

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ ls
Finance HR IT

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ |
```

Question 20: Comprehensive Linux Administrator Task

Perform the following tasks related to comprehensive linux administrator task:

1. Create a directory named linux_exam.
2. Create users.txt, logs.txt and backup.sh.
3. Assign execute permission to backup.sh.
4. Search for the word Linux in users.txt.
5. Display inode numbers.
6. Create a compressed archive.
7. Display disk usage.
8. Display memory usage.

9. Save outputs into admin_report.txt.

Submit: Screenshot showing commands executed and corresponding outputs.

```
/cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ mkdir linux_exam

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ cd linux_exam/

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ touch users.txt logs.txt backup.sh

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ ls -l
total 0
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 11:45 backup.sh
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 11:45 logs.txt
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 11:45 users.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ chmod +x backup.sh

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ ls -l backup.sh
-rwxrwx--x+ 1 Tavakalmastan Tavakalmastan 0 Jun 16 11:45 backup.sh

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ echo "Linux" > users.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ grep "Linux" users.txt
Linux

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ ls -li
281474977294918 backup.sh 281474977294916 users.txt
281474977294917 logs.txt

Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
```

 /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company/linux_exam
$ cd ../
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ tar -czvf linux_exam.tar.gz users.txt logs.txt backup.sh
tar: users.txt: Cannot stat: No such file or directory
tar: logs.txt: Cannot stat: No such file or directory
tar: backup.sh: Cannot stat: No such file or directory
tar: Exiting with failure status due to previous errors
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ ls
Finance  HR  IT  linux_exam  linux_exam.tar.gz
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ ls -l linux_exam.tar.gz
-rwxrw----+ 1 Tavakalmastan Tavakalmastan 45 Jun 16 11:48 linux_exam.tar.gz
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ du -sh
2.0K  .
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ du -sh .
2.0K  .
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ systeminfo | findstr /C:"Total Physical Memory"
```

```
Total Physical Memory:          16,069 MB
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ systeminfo | findstr /C:"Total Physical Memory" >> admin_report.txt
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$ cat admin_report.txt
Total Physical Memory:          16,069 MB
```

```
Tavakalmastan@Mastan /cygdrive/E/GVIM_practice/pd/PD_training/linux/employees/project/company
$
```

***** END OF QUESTION PAPER *****

